Table 9.1: Different types of tissues					
	Distribution	**Main functions**	**Nature**	**Cell shape**	**Wall materials**
Parenchyma	Cortex, Pith medullary rays and Packing tissues in vascular system	Packing tissue, support, gaseous exchange, food storage	Living	Usually Isodiametric	Mainly Cellulose and Pectin
Collenchyma	Outer region of cortex as in angles of stems, mid-rib of leaves	Mechanical	Living	Elongated, Polygonal	Mainly Cellulose, Pectin and Hemi-cellulose
Sclerenchyma (a) Fibre	Outer region of cortex, pericycle of stems, vascular bundles	Mechanical	Dead	Elongated and Polygonal with tapering ends	Mainly Lignin
(b) Sclereids	Cortex, Pith, Phloem shells and stones of fruits and seed coats	Mechanical Protection	Dead	Roughly Isodiametric with much variation	Mainly lignin
Tracheids and Vessels	Vascular System	Translocation of water and mineral salts	Dead	Elongated and Tubular	Mainly lignin
Phloem Sieve tubes	Vascular System	Translocation of organic solutes	Living	Elongated and Tubular	Cellulose, Pectin and Hemicellulose
Companion Cells	Vascular System	Work in association with sieve tubes	Living	Elongated and narrow	Cellulose, Pectin and Hemicellulose

Difference Between Meristematic Tissue and Permanent Tissue

Meristematic tissue	Permanent tissue
• Cells divide repeatedly • Cells are undifferentiated • Cells are small and Isodiametric • Intercellular spaces are absent • Vacuoles are absent • Cell walls are thin • Inorganic inclusions are absent	• Do not divide • Cells are fully differentiated • Cells are variable in shape and size • Intercellular spaces are present • Vacuoles are present • Cell walls maybe thick or thin • Inorganic inclusions are present

Difference Between Collenchyma and Sclerenchyma

Collenchyma	Sclerenchyma
• Living Cells • Contains Protoplasm • Cell walls are cellulosic • Thickening of cell wall is not uniform • Keeps the plant body soft • Sometimes it has chloroplast	• Dead cells • Do not have protoplasm • Cell walls are lignified • Thickening of cell wall is uniform • Keeps plant body stiff and hard • Do not have chloroplast

XIth_Botany_Chapter_9_EM.indd 204 5/17/2020 9:16:38 PM